Physics 9702 Paper 4 · Topic 20

Magnetic fields

The seven hardest A Level Structured Questions on this topic from the 2020 to 2026 papers, hardest first, followed by their mark schemes.

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1 9702/44 May/June 2025, Question 6 (10 marks) page 2

Examiner report: Weaker candidates found more difficulty in identifying which side of the coil was being referred to as L in their equation. (iii) Only the strongest candidates were able to sketch the sinusoidal curve without error, despite some allowance being given for a freehand drawing.

2 9702/44 May/June 2025, Question 7 (8 marks) page 4

Examiner report: (iii) This was one of the more difficult questions on the paper. Only the strongest candidates were able to do so, with many candidates focusing instead on the impact of Lenz's law and opposing forces.

3 9702/42 May/June 2022, Question 6 (10 marks) page 6

Examiner report: Weaker candidates found it difficult to articulate some of this, with some suggesting that the wire had to be able to carry current (i.e. be a conductor) or giving information that was already provided in the question stem. The weakest candidates confused the situation with electromagnetic induction and thought the wire had to be moving, or confused carrying a current with the wire being charged.

4 9702/41 May/June 2021, Question 10 (9 marks) page 8

Examiner report: (b) Candidates found this question difficult. Many candidates did not realise the importance of the current and found it difficult to set out the logical steps needed to explain what is happening.

5 9702/42 May/June 2022, Question 7 (10 marks) page 10

Examiner report: The most difficult aspect was to realise that the magnitude of the induced e.m.f. in the middle section must be three times the magnitude before t = 0.15 s and after t = 0.25 s. Of those who did realise it would still be upwards, very few candidates were able to give a convincing explanation.

6 9702/42 February/March 2024, Question 6 (11 marks) page 12

Examiner report: (ii) Very few candidates were able to explain the constancy of flux linkage in terms of the magnetic field having a constant flux density inside the solenoid. The most difficult of the three marking points was the one that required a discussion of how the magnetic fields due to adjacent turns interact with each other to cause a force between the turns.

7 9702/42 October/November 2024, Question 1 (13 marks) page 15

Examiner report: Confusion with Faraday's law was common, with many candidates giving responses in terms of the magnitude of the induced e.m.f. (v) This was a difficult question, with only the very strongest candidates demonstrating an ability to use Lenz's law correctly.

Questions were chosen using the Cambridge examiner reports (Principal Examiner Report for Teachers): parts the reports describe as challenging, poorly answered, or answered fully only by the strongest candidates. May/June 2020 and May/June 2026 have no report, so questions from those series were judged directly. Variant 43 repeats variant 41 in every series, so only 41 is cited.

6 A rectangular coil PQRS of wire is free to rotate about its axis XY, as shown in Fig. 6.1.

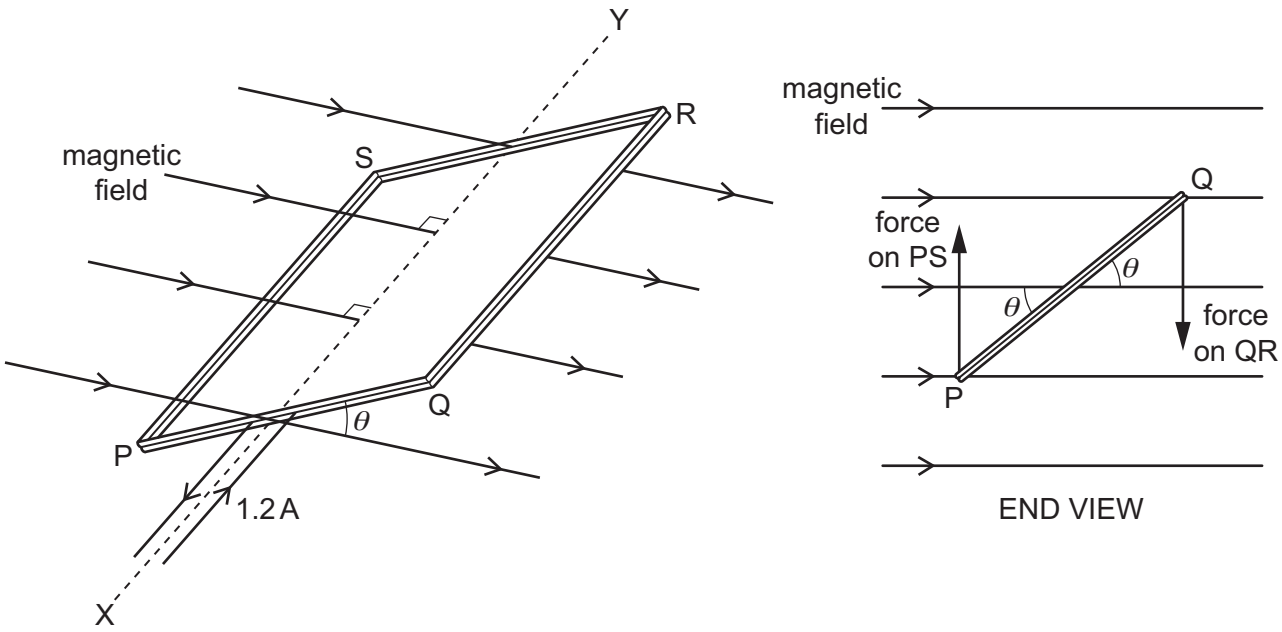


Fig. 6.1 (not to scale)

The coil has length QR of 5.4 cm, width PQ of 2.5 cm and has 190 turns of wire.
The plane of the coil is at an angle θ to a uniform magnetic field of flux density 5.2×10^{-3} T.
The axis XY of the coil is normal to the field.
The current in the coil is 1.2 A.

(a) (i) Calculate the magnitude of the force on side QR of the coil.

force = ... N [3]

(ii) Use your answer in **(a)(i)** to show that the torque τ on the coil is given by

$$\tau = 1.6 \times 10^{-3} \cos\theta \text{ N m}.$$

[2]

(iii) Using the expression in **(a)(ii)** sketch, on the axes of Fig. 6.2, a graph to show the variation of the torque τ with angle θ for values of θ between 0 and 360°. Label the τ axis with an appropriate scale.

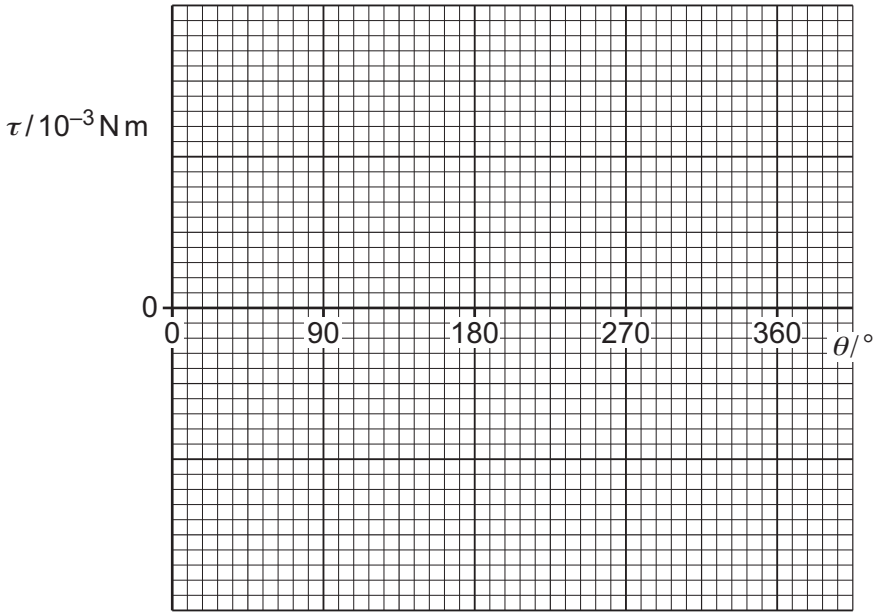


Fig. 6.2

[3]

(b) The coil is now replaced by an identical coil wound on a ferrous core.

Suggest, with a reason, how the torque on this coil compares with the torque on the original coil.

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..... [2]

[Total: 10]

7 A bar magnet is suspended from a spring. One pole of the magnet oscillates freely in a coil of wire, as shown in Fig. 7.1.

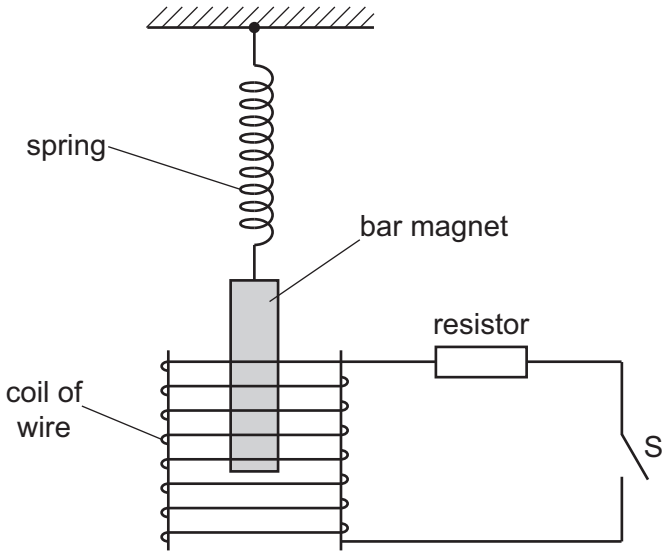


Fig. 7.1

The switch S is initially open.

(a) The switch S is now closed. As a result, the oscillations of the magnet are lightly damped.

(i) State what is meant by damping.

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.....

..... [2]

(ii) Describe what is observed to indicate that the damping is light.

.....

..... [1]

(iii) By reference to electromagnetic induction and to conservation of energy, explain why the oscillations are damped.

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.....

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..... [3]

(b) The procedure in **(a)** is repeated after replacing the resistor with one of greater resistance.

Suggest, with a reason, the effect of this change on the oscillations.

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.....

..... [2]

[Total: 8]

6 **(a)** State the **two** conditions that must be satisfied for a copper wire, placed in a magnetic field, to experience a magnetic force.

1 .....

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2 .....

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[2]

(b) A long air-cored solenoid is connected to a power supply, so that the solenoid creates a magnetic field. Fig. 6.1 shows a cross-section through the middle of the solenoid.

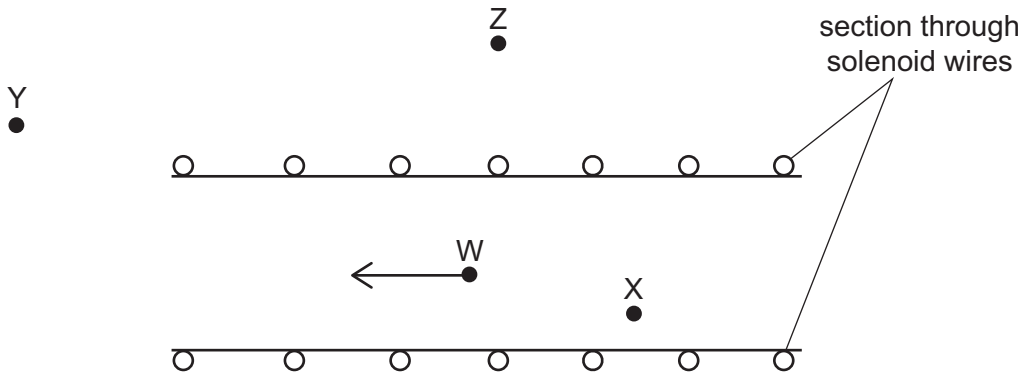


Fig. 6.1

The direction of the magnetic field at point W is indicated by the arrow. Three other points are labelled X, Y and Z.

(i) On Fig. 6.1, draw arrows to indicate the direction of the magnetic field at each of the points X, Y and Z. [3]

(ii) Compare the magnitude of the flux density of the magnetic field:

- at X and at W .....

.....

- at Y and at Z. .....

.....

[2]

(c) Two long parallel current-carrying wires are placed near to each other in a vacuum.

Explain why these wires exert a magnetic force on each other. You may draw a labelled diagram if you wish.

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..... [3]

[Total: 10]

10 (a) State Lenz's law.

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..... [2]

(b) A metal ring is suspended from a fixed point P by means of a thread, as shown in Fig. 10.1.

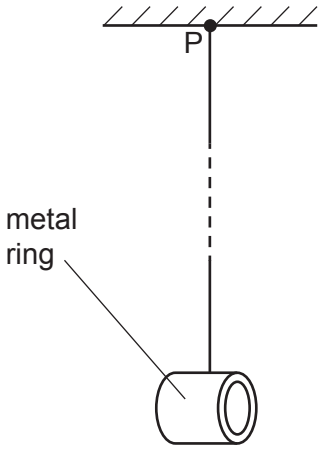


Fig. 10.1

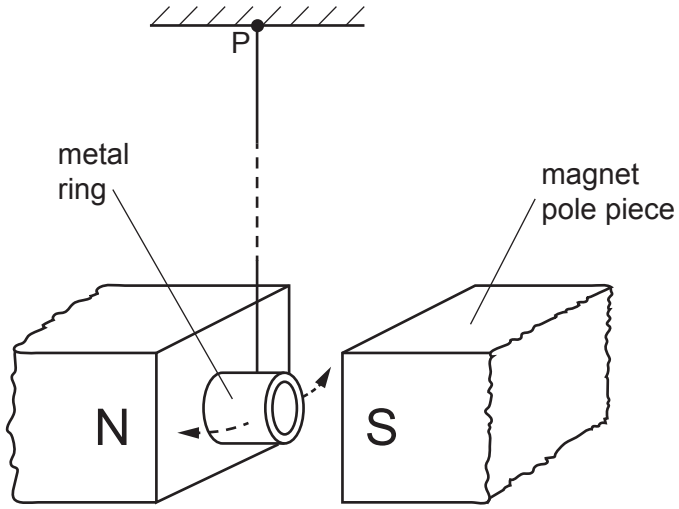


Fig. 10.2

The ring is displaced a distance *d* and then released. The ring completes many oscillations before coming to rest.

The poles of a magnet are now placed near to the ring so that the ring hangs midway between the poles of the magnet, as shown in Fig. 10.2.

The ring is again displaced a distance *d* and then released.
Explain why the ring completes fewer oscillations before coming to rest.

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.....

..... [4]

(c) The ring in **(b)** is now cut so that it has the shape shown in Fig. 10.3.



Fig. 10.3

Explain why, when the procedure in **(b)** is repeated, the cut ring completes more oscillations than the complete ring when oscillating between the poles of the magnet.

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..... [3]

[Total: 9]

7 **(a)** State Faraday's law of electromagnetic induction.

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..... [2]

(b) Two coils are wound on an iron bar, as shown in Fig. 7.1.

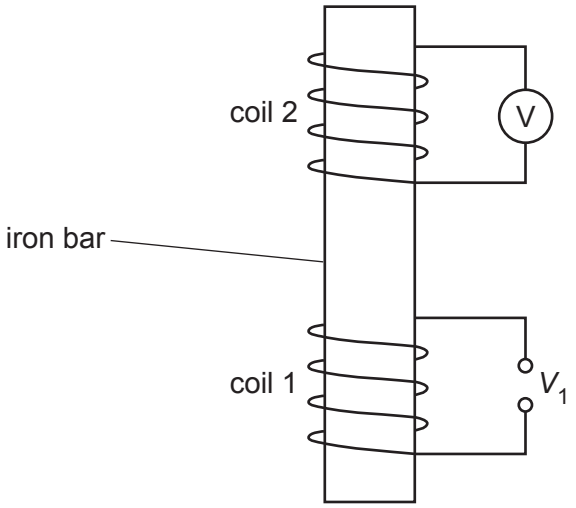


Fig. 7.1

Coil 1 is connected to a potential difference (p.d.) V_1 that gives rise to a magnetic field in the iron bar.

Fig. 7.2 shows the variation with time t of the magnetic flux density B in the iron bar.

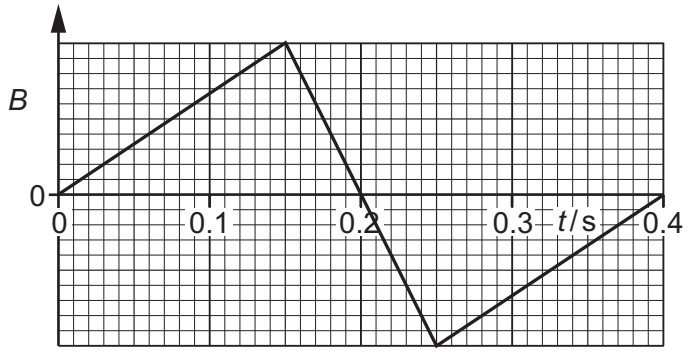


Fig. 7.2

A voltmeter measures the electromotive force (e.m.f.) V_2 that is induced across coil 2.

On Fig. 7.3, sketch the variation with t of V_2 between $t = 0$ and $t = 0.40$ s.

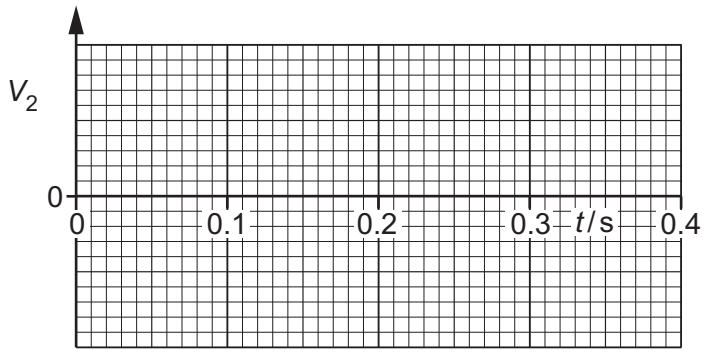


Fig. 7.3

[4]

(c) Coil 2 in **(b)** is now replaced with a copper ring that rests loosely on top of coil 1. The supply to coil 1 is replaced with a cell and a switch that is initially open, as shown in Fig. 7.4.

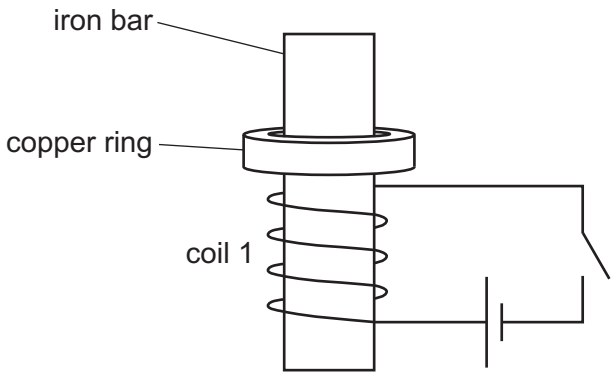


Fig. 7.4

(i) The switch is now closed. As it is closed, the copper ring is observed to jump upwards.

Explain why this happens.

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.....

..... [3]

(ii) Suggest, with a reason, what would be the effect of repeating the procedure in **(c)(i)** with the terminals of the cell reversed.

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..... [1]

[Total: 10]

6 **(a)** A small coil C has 64 turns and cross-sectional area 0.71 cm^2. The coil is placed inside a solenoid as shown in Fig. 6.1.

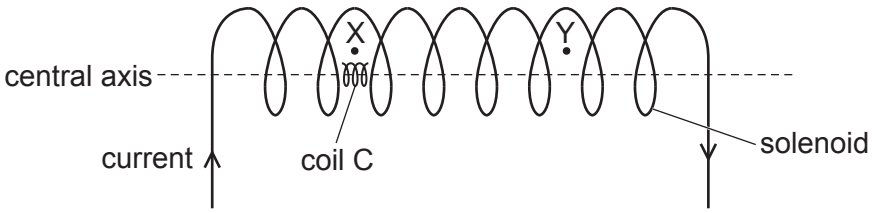


Fig. 6.1

The centre of coil C is on the central axis of the solenoid.

(i) There is a constant current in the solenoid.
Coil C is moved through the solenoid from position X to position Y.

On Fig. 6.2, sketch a line to show the variation of the magnetic flux linkage in coil C with position as it moves from X to Y.

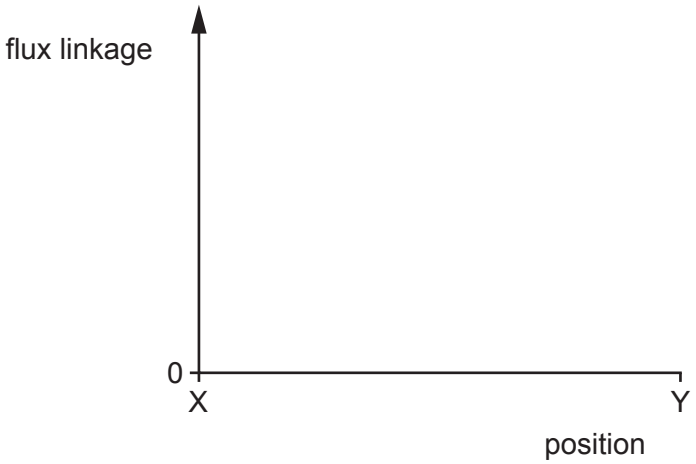


Fig. 6.2

[1]

(ii) Explain the shape of your line in **(a)(i)**.

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.....

..... [2]

(iii) Coil C is now held stationary at X. The current in the solenoid varies so that the magnetic flux density *B* at X varies from time 0 to time 4*t* as shown in Fig. 6.3.

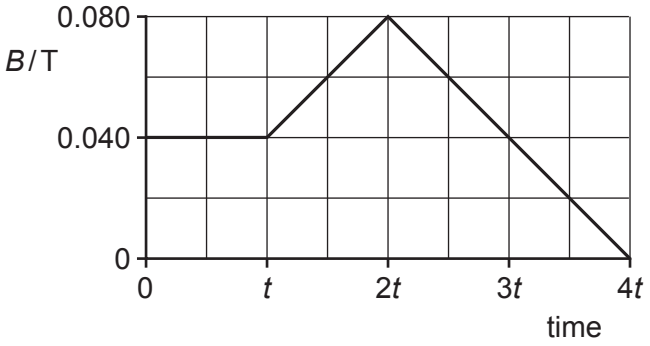


Fig. 6.3

Calculate the maximum magnetic flux linkage in coil C.

flux linkage = ... Wb [2]

(iv) On Fig. 6.4, sketch a line to show the induced electromotive force (e.m.f.) *E* in coil C from time 0 to time 4*t*.

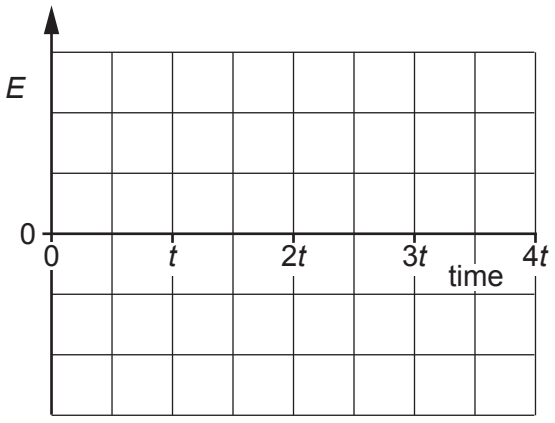


Fig. 6.4

[3]

(b) A metal spring rests on a smooth table. The turns of the spring are equally spaced. The ends of the spring are connected to a d.c. power supply, as shown in Fig. 6.5.

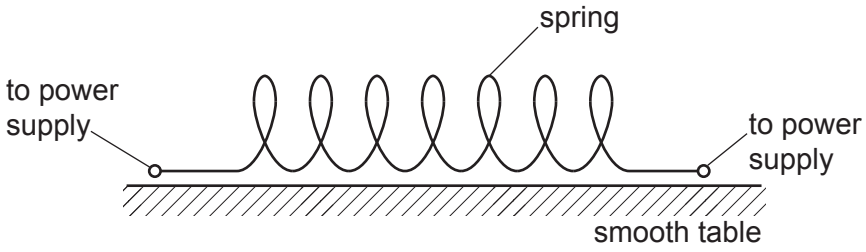


Fig. 6.5

The spring is connected to the d.c. power supply using flexible leads. The spring is not under tension.

With reference to magnetic fields, describe and explain the change in the distance between the turns of the spring when the power supply is first switched on.

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..... [3]

[Total: 11]

1 A metal wheel consists of an axle A, eight spokes and a rim, as shown in Fig. 1.1.

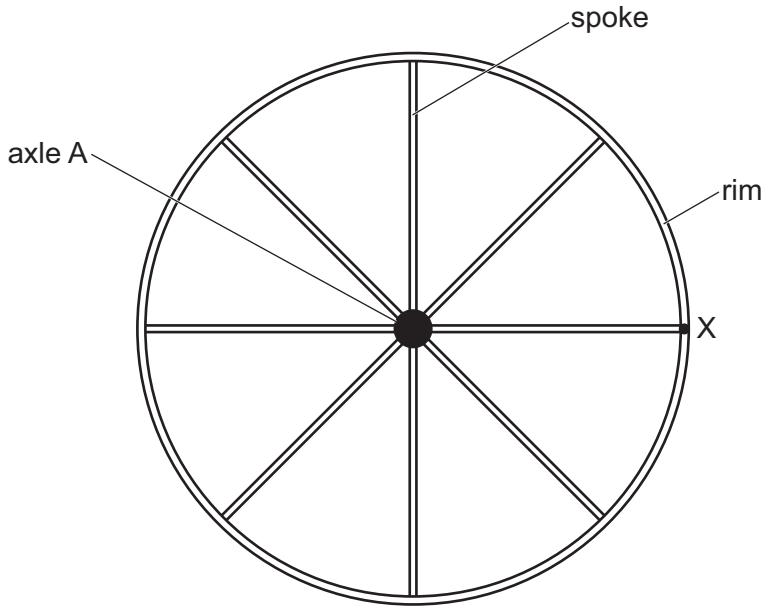


Fig. 1.1

Point X is on the rim at the end of one of the spokes.

The rim has a radius of 0.85 m.
The wheel is rotating clockwise with an angular speed of 140 rad s^{-1}.

(a) For point X, determine:

(i) the speed

speed = ... $m\,s^{-1}$ [2]

(ii) the centripetal acceleration.

acceleration = ... $m\,s^{-2}$ [2]

(b) There is a uniform magnetic field of flux density 0.18 T into the plane of the page.

(i) State Lenz's law of electromagnetic induction.

.....

.....

..... [2]

(ii) Show that the time taken for point X to complete one revolution is 45 ms.

[1]

(iii) Calculate the magnetic flux cut by spoke AX during one revolution of the wheel.
Give a unit with your answer.

magnetic flux = ... unit ..... [3]

(iv) Determine the magnitude of the electromotive force (e.m.f.) induced across spoke AX.

induced e.m.f. = ... V [2]

(v) Use Lenz's law to explain whether the potential is higher at end A or end X of the spoke.

.....

.....

..... [1]

[Total: 13]

Question	Answer	Marks
6(a)(i)	$F = BIL$	**C1**
	force on QR = $5.2 \times 10^{-3} \times 1.2 \times 0.054 \times 190$	**C1**
	= 0.064 N	**A1**
6(a)(ii)	torque = force × perpendicular distance between forces	**C1**
	= $0.064 \times 0.025 \cos \theta = (1.6 \times 10^{-3}) \cos \theta$ N m	**A1**
6(a)(iii)	one complete cycle of a sinusoidal curve between 0 and 360°	**B1**
	τ axis labelled to show maximum and minimum torques at ± 1.6×10^{-3} N m	**B1**
	maximum at 0 and 360° and minimum at 180° (or vice versa), with torque shown as zero at 90° and 270°	**B1**
6(b)	(ferrous core) increases magnetic flux density	**B1**
	amplitude of torque increases	**B1**

Question	Answer	Marks
7(a)(i)	loss of energy (of oscillations)	**B1**
	due to resistive forces	**B1**
7(a)(ii)	amplitude (of oscillations) decreases gradually **or** oscillations continue (for several periods)	**B1**
7(a)(iii)	cutting of (magnetic) flux causes induced e.m.f. (in coil) **or** induced e.m.f. causes current (in resistor / circuit)	**B1**
	current causes dissipation of thermal energy in resistor **or** current in resistor causes dissipation of thermal energy	**B1**
	thermal energy comes from energy of oscillations	**B1**
7(b)	(for any given e.m.f.) current is lower so thermal energy dissipated at a lower rate **or** (for any given e.m.f.) current is lower so the resistive force is lower	**B1**
	oscillations are less damped **or** *s*maller decrease in amplitude of oscillations (in each period) **or** oscillations continue for longer	**B1**

Question	Answer	Marks
6(a)	there must be a current (in the wire)	**B1**
	(wire) must be at a non-zero angle to the magnetic field	**B1**
6(b)(i)	arrow from X pointing horizontally to the left	**B1**
	arrow from Y pointing diagonally upwards and to the left at about 45°	**B1**
	arrow from Z pointing horizontally to the right	**B1**
6(b)(ii)	(flux densities at W and X are approximately) equal	**B1**
	(flux density at) Y greater than (flux density at) Z	**B1**
6(c)	current in wire creates magnetic field around wire	**B1**
	(each) wire sits in the magnetic field created by the other	**B1**
	(for each wire,) current / wire is perpendicular to magnetic field (due to other wire), (so) experiences a (magnetic) force	**B1**

Question	Answer	Marks
10(a)	direction of (induced) e.m.f.	**M1**
	is such as to oppose the change causing it	**A1**
10(b)	ring cuts (magnetic) flux and causes induced e.m.f. in ring	**B1**
	(induced) e.m.f. causes (eddy/induced) currents (in ring)	**B1**
	currents (in ring) cause magnetic field (around ring)	**M1**
	two fields interact to cause resistive/opposing force	**A1**
	or	
	current (in ring) is in a magnetic field	**(M1)**
	which causes resistive force	**(A1)**
	or	
	currents (in ring) dissipate thermal energy	**(M1)**
	(thermal) energy comes from energy of oscillations	**(A1)**
10(c)	current cannot pass all the way around the ring	**B1**
	(induced) currents smaller	**B1**
	smaller resistive force (so more oscillations) **or** smaller rate of dissipation of energy (so more oscillations)	**B1**

Question	Answer	Marks
7(a)	induced e.m.f. is (directly) proportional to rate	**M1**
	of change of (magnetic) flux (linkage)	**A1**
7(b)	V_2 stepped, all at non-zero values, between t = 0 and t = 0.40 s	**B1**
	V_2 shown with same non-zero magnitude up to t = 0.15 s and after t = 0.25 s but with a different magnitude between these times	**B1**
	V_2 shown with a magnitude between t = 0.15 s and t = 0.25 s that is three times the magnitude before t = 0.15 s and after t = 0.25 s	**B1**
	V_2 shown with same sign up to t = 0.15 s and after t = 0.25 s, and opposite sign in between	**B1**
7(c)(i)	changing current in coil causes changing (magnetic) field **or** changing (magnetic) flux causes induced e.m.f. in ring	**B1**
	induced e.m.f. in ring causes current in ring	**B1**
	(magnetic) field due to (induced) current in ring interacts with (coil's) field to cause upwards force (on ring) **or** (induced) current in ring perpendicular to (coil's magnetic) field causes upwards force (on ring)	**B1**
7(c)(ii)	both magnetic fields reverse direction so ring still jumps up **or** current (in ring) and (coil's) field both reverse so ring still jumps up	**B1**

Question	Answer	Marks
6(a)(i)	non-zero horizontal straight line from X to Y	**B1**
6(a)(ii)	constant flux density (inside coil)	**B1**
	either (magnetic) flux linkage proportional to flux density **or** $\Phi = BAN$ and B, A and N are all constant	**B1**
6(a)(iii)	$\Phi = BAN$	**C1**
	$= 0.080 \times 0.71 \times 10^{-4} \times 64$ $= 3.6 \times 10^{-4}$ Wb	**A1**
6(a)(iv)	*sketch showing*: E is zero from time 0 to time t and non-zero after time t	**B1**
	E has constant non-zero magnitude between time t and time $4t$	**B1**
	E has non-zero value of one sign between time t and time $2t$, and non-zero value of the opposite sign between time $2t$ and time $4t$	**B1**
6(b)	current in spring creates a magnetic field around the spring	**B1**
	either (magnetic) fields around adjacent turns interact to cause a force to be exerted (between the turns) **or** current in one turn interacts with (magnetic) field due to adjacent turns to cause force to be exerted (between the turns)	**B1**
	(magnetic force) is attractive so distance (between turns) decreases	**B1**

Question	Answer	Marks
1(a)(i)	$v = r\omega$	**C1**
	$= 0.85 \times 140$ $= 120\ \text{m s}^{-1}$	**A1**
1(a)(ii)	$a = r\omega^2$ **or** $a = v^2 / r$	**C1**
	$a = 0.85 \times 140^2$ **or** $120^2 / 0.85$ $= 1.7 \times 10^4\ \text{m s}^{-2}$	**A1**
1(b)(i)	direction of (induced) e.m.f.	**M1**
	is such as to (produce effects that) oppose the change that caused it	**A1**
1(b)(ii)	$T = 2\pi / \omega$ $= 2\pi / 140 = 0.045\ \text{s} = 45\ \text{ms}$	**A1**
1(b)(iii)	$\Phi = BA$	**C1**
	$= 0.18 \times \pi \times 0.85^2$	**C1**
	$= 0.41\ \text{Wb}$	**A1**
1(b)(iv)	$E = \Phi / t$	**C1**
	$= 0.41 / 0.045$ $= 9.1\ \text{V}$	**A1**
1(b)(v)	force (on spoke) must be anticlockwise, so current is from A to X (by Fleming's left hand rule), so X is at the higher potential	**B1**